Fiscal Year:	FY 2005	**Task Last Updated:**	FY 04/03/2006
PI Name:	Schlegel, Robert E. Ph.D.		
Project Title:	Development and Evaluation of Remote Manipulator System Operator Proficiency and Training Effectiveness Metrics		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Space human factors engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Norman	**State:**	OK
Zip Code:	73019	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	01-OBPR-07
Start Date:	04/01/2003	**End Date:**	03/31/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: funding delayed and project extended until 3/31/2007 per PI (4/06 info added 4/08--Ed.)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG 9 1486		
Performance Goal No.:			
Performance Goal Text:			
	Remote manipulator systems (RMS) exist aboard the International Space Station and the Space Shuttle to assist in space operations such as material management, transportation, unberthing, retrieval and berthing of payloads, and construction tasks. Each RMS is remotely controlled by an astronaut using two hand controllers (one translational and one rotational) and an interface panel while monitoring robot arm movement on closed circuit television displays. Prior to launch, astronauts undergo several phases of training to acquire the skill set necessary for successful operation of the arms. The early phases of training involve the use of generic dynamic skills training simulators. Although the training program offers instructors objective criteria for evaluating trainee progress, there are currently no quantitative operator performance data generated or recorded to aid instructors in their evaluations. The objective of the current research is to identify and validate relevant quantitative measures of RMS operator proficiency in a format that enables instructors, trainees, and trained operators to appraise current skill levels and to		

Task Description:	determine candidate areas for additional training or practice. The research will likewise identify metrics that can serve as in-flight readiness-to-perform indicators prior to critical operational tasks. The research will begin by augmenting the current operator proficiency criteria with additional metrics used to assess teleoperator performance obtained from a review of literature, task analysis, and interviews with trainers and trainees. The research will proceed by recording operator activity data and robot arm movement data from a number of participants at various stages of the training process. Instructor ratings will be used to confirm differing levels of expertise along the training progression of the participants. Statistical analyses will be used to isolate the measures of operator activity and robot arm movement that provide the best discriminability between expert and non-expert operators. Finally, displays of the selected measures will be developed to aid instructors and trainees in assessing training progress and skill proficiency. Following incorporation into the training program, the methodology will be validated against the present system of instructor evaluation of trainees. Future extensions of the work would focus on system- and mission-specific implementations of the metrics and application in a readiness-to-perform setting in flight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our current NASA research forms a component of a larger program in human performance assessment at the University of Oklahoma within the Center for the Study of Human Operator Performance. Successful development and validation of the RMS metrics will demonstrate a procedure that can be applied in other performance assessment areas, both within and outside the realm of human telerobotics control. This research will also lead to improved cognitive performance assessment in military and commercial contexts.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/03/2020)
Articles in Peer-reviewed Journals	Bray TM, Shehab RL, Schlegel RE, Civan A, Walker DE. "Effectiveness of 2-D views for 6-D robotics simulation maneuvers." Proceedings of the Human Factors and Ergonomics Society Annual Meeting. 2003 Oct;47(8):975-9. (47th Annual Meeting of the Human Factors and Ergonomics Society, Denver, Colorado, October 13 - 17, 2003.) https://doi.org/10.1177/154193120304700805 , Oct-2003